CHAPTER-1

INTRODUCTION

1.1 GENERAL INTRODUCTION

The Indian astrological calendar known as Panchangam plays an important role in the daily lives of all Indians and is part of their cultural heritage. Panchangams are used to determine when to do important activities such as: travel, do business, perform religious rituals, and celebrate festivals. The Panchangam contains a lot of information which includes, but is not limited to: sunrise, sunset, Rahu Kalam, Yamagandam, Gulikai, and other various periods of time that are significant in deciding when to conduct activities.

With printed calendars or online websites being the primary means by which most people access Panchangam, there are limitations (i.e., printed calendars are static with no ability to reflect real-time changes and the need for an internet connection on online systems). These limitations reduce the functionality and capability of traditional Panchangam.

Advancements in both electronics and embedded systems have provided the ability to create compact and intelligent systems that can calculate data in real-time while creating the ability to develop an Edge AI-Driven Electronic Indian Astrological Calendar that is able to perform accurately without requiring an internet connection.

The system contains an Arduino microprocessor, Real-Time Clock (RTC), TFT display and a sensor array to calculate and display astrological data. By using traditional knowledge with modern technology, the efficiency of this system is improved by making it lightweight, portable and easy to use.

This also reduces the labour involved in calculating the times manually by automating the processes thereby increasing the accuracy of the output. Additionally, this system makes it easier for users to access astrological information that may not have access to traditional Panchangam formats. The digital representation of this data helps to provide a user-friendly solution for both younger people, who tend to use electronic devices more than printed materials, as well as older generations who are not comfortable with using computers.

1.2 NEED FOR THE PROPOSAL

There are many reasons to develop this project that examine the challenges of current techniques to retrieve astrological content. First, printed Panchangams do not include real-time updates. Users must interpret the information themselves, which can sometimes lead to confusion or error in interpretation.

Second, many online solutions for accessing astrological content require an active internet connection. This often creates a lot of frustration for users located in rural areas or without consistent service during ceremonial or significant occasions.

Third, there aren't small portable devices available to provide users with quick access to all astrological content offline. A need exists for a simple, reliable method of accessing astrological content anytime and from anywhere.

This project will provide a stand-alone embedded system for delivering current and accurate astrological content without requiring any internet service to access the content. The use of Edge Artificial Intelligence will allow for increased speed, reliability and performance of the installation capabilities.

Finally, with the growing use of smartphones and other smart devices to conduct daily activities, the likelihood that individuals will prefer digital solutions to support traditional industries continues to grow.

1.3 APPLICATIONS

A variety of uses for the proposed system exist in both everyday and religious or educational spheres, including,

- Calculating daily astrological timings such as sunrise, sunset, Rahu Kalam, Yamagandam, and Gulikai;
- Planning important tasks and events according to auspicious times (for example: travel, business).

- Scheduling pooja and other ceremonies in temples and other religious venues.
- Providing education on how to read the Panchangam and perform basic astronomical calculations.
- Assisting users who live in areas without reliable internet access.
- Enabling users to travel and carry this device wherever they go as portable equipment.
- Combining traditional knowledge with contemporary embedded systems technology.

Astrologers and anyone else researching classical Indian time perceptions can benefit from using this system; it also provides daily notifications/alerts regarding good times for individual and smart home/IoT devices (better using).

Also users can define their preferences for notification/alert settings by user (like daily routine) including time & place; therefore having 1 place used repeatedly will keep consistency in rituals & schedules (no longer manual). Using these types of contemporary technologies allows for quicker processing of data and providing an accurate way of doing things; this means the potential to easily help adapt.

1.4 ORGANIZATION OF THE CHAPTER

Chapter Sections per Sections

1.1 - Introduction to the Indian Astrology Calendar

1.2 – Justification for Implementation of the New System

1.3 – All Project Goals

1.4 - How to Use the New System

Summary of the Chapter that will highlight key points emphasising all chapters.

Each section in this chapter of the project will aid the readers in having a better identify of the purpose/scope of this project.

Each section in each chapter of the project will provide the reader a way of proceeding from one point to another sequentially. There was a reasonable logical connection among the sections in all chapters of the project, thus helping readers get a good understanding of the problem being solved through new development, projected solution, and projected outcome. Therefore making the contents of this project easily understood and increasing the usefulness of this project to fulfil the purposes of academic review.

1.5 CONCLUSION

In this chapter, we will discuss the Indian astrological calendar and its relevance in society. The limitations of traditional versus online approaches indicate a real demand for an alternative solution to overcome these challenges.

The solution proposed herein is a portable, reliable and offline, Edge AI (Artificial Intelligence)-based electronic Panchangam, which will provide users with access to astrological data using Embedded Systems/Deep Learning technology.

This chapter aims to lay the foundation for the following chapters describing the design, implementation and results of the proposed system.

In addition to providing a way to utilize modern technology to preserve traditional wisdom digitally, this project seeks to create a way to preserve cultural practices while finding an efficient method to share those practices across multiple regions. The system is likely to benefit many users as well as open up new avenues for developing more efficient systems and pursuing additional research in this area.

Moreover, the system will be scalable and adaptable, allowing future enhancements such as integration with mobile applications and cloud service providers. It is also anticipated that by supporting multiple regional languages, the system will make it more accessible to diverse groups of users. Improved usability and adaptability increase the likelihood of widespread acceptance and successful deployment in real-world applications.

CHAPTER-2

LITERATURE REVIEW

2.1 INTRODUCTION

The traditional method of relying on almanacs and cloud-based platforms for Panchangam information—a vital astrological tool for Indian culture and daily life—is challenged by a critical bottleneck: the "connectivity gap." Although traditional almanacs are non-connective, cloud-based solutions require a continuous internet connection for information on dynamic astrological data such as Rahu Kalam or solar cycles.

In response to these challenges, this research aims to revolutionize astrological information systems based on a paradigm shift to Edge AI technology. With the integration of an Arduino microprocessor and a Real-Time Clock (RTC), computations are shifted directly to the 'edge' of the network.

Zero Latency Autonomy: Complex computations are carried out locally within milliseconds, eliminating latency for instant results. Network Independance: The astrological information system can function independently of internet connectivity, providing unfettered access to traditional knowledge systems. Improved Data Integrity: With computations carried out on-site, there is no risk of data compromise or hacking while transmitting information via external networks.

This Edge AI technology is not only capable of automating complex manual calculations with high temporal accuracy but can also become a user-friendly interface between traditional knowledge systems and contemporary embedded technology.

2.2 KEY PARAMETERS DISCUSSION

Zambetti, N. [1] The documentation provides detailed technical specifications of the Arduino Nano , which can be used in embedded systems, including its architecture, processor type, clock frequency, memory capacity and input/output pin configuration. The small size and low power consumption of the Arduino Nano make it an ideal choice for compact designs; additionally, since it's compatible with many different communication protocols (e.g. UART, SPI, I2C) , multiple devices (sensors, displays, peripherals, etc.) can easily be connected via I2C – which greatly reduces the complexity of wiring and therefore increases the efficiency of the system. Moreover, Arduino Nano is compatible with the Arduino IDE, which simplifies programming for users without prior experience. Due to its reliability and flexibility, Arduino Nano is frequently found in both real-time applications and Internet of Things (IoT) devices. In conclusion, based on all the information provided in this document, we have shown that the Arduino Nano is a very effective platform for developing portable and standalone embedded systems.

Banzi, M., & Cuartielles, D. [2] The documentation describes the usage of the Wire library, a software library for Arduino base systems that support I2C (Inter-Integrated Circuit) communications. The Wire library allows you to connect multiple devices (e.g., sensors, real-time clocks (RTC), and LCDs) to the microcontroller using only two wires (SDA and SCL), thus greatly reducing the complexity of the wiring within your system and decreasing the overall amount of connectors that you will need to manage within your hardware. The Wire library also facilitates communications between master and slave devices without any interference. Additionally, the Wire library contains several methods to initialize communications, send data, receive responses and handle other aspects of working with an I2C network. In embedded systems that require multiple components to communicate seamlessly, this library is essential. It simplifies the way these systems communicate, making it easier for developers to program their applications and improving overall system functioning. In addition, the Wire library is vital for enhancing connectivity, reliability, and integration in contemporary embedded systems design.

Fried, L. [3] Maxim Integrated (2021) The DS3231 Real-Time Clock (RTC) Module - Maxim Integrated (2021). This document is a data sheet providing information on the DS3231 RTC Module, which provides a very Accurate and reliable timekeeping solution. It provides precision timekeeping via the use of temperature compensated crystal oscillators (TCXO). These TCXO allow the DS3231 to keep accurate time over long periods, while also allowing it to continue keeping accurate time even after power has been removed by using an onboard battery backup mechanism. The DS3231 is designed to communicate with microcontrollers using the I2C protocol, thus allowing developers to easily incorporate it into their Arduino projects. Additionally, the DS3231 consumes very little power and provides higher precision when compared to other RTC devices. Because of all of these attributes, the DS3231 may be commonly utilized in applications requiring highly accurate timing, such as scheduling systems, and/or astrological calculations.

Mellis, D. [4] This datasheet provides details about BMP280 digital pressure sensor which is capable to measure atmospheric pressure, temperature and altitude measurements. BMP280 is extensively used in embedded systems environmental monitoring applications. BMP280 sensor has a high level of accuracy, high-speed response time, allowing real-time data acquisition and it supports both I2C and SPI communications providing the user an option for implementing the sensor on a variety of microcontrollers. BMP280 also has a very low power consumption and a very small physical form factor, thus making it very well suited for portable and battery powered devices. BMP280 will provide real-time environmental data thus improving the accuracy of calculations used in various applications. Overall, BMP280 is an excellent and efficient sensor for the next generation of embedded systems.

Hareendran, T.K. [5] You can get reliable location information using the NEO-6M GPS module through GPS, GLONASS and GALILEO satellite networks. The NEO-6M Provides a precise Position using latitude coordinates, longitude coordinates and altitude coordinates, making it an ideal tool for Navigation and Vehicle Tracking. The NEO-6M GPS Receiver works with all major satellites (GPS, GLONASS and GALILEO) and provides the means for electrical components to achieve reliable navigation and vehicle tracking to the end-user. The communication between the NEO-6M Receiver and electrical components occurs by means of Serial Communications Protocols; this makes it easy to integrate within an embedded application. In addition to serial interface availability, the data sheet provides quick position measurement characteristics as well as low power consumption for battery-powered products that require little maintenance. Incorporating GPS into electrical devices improves the accuracy and reliability of geographic position calculations performed by those devices. This provides increased functionality, accuracy, and precision to electrical devices for the end user. Therefore, we believe the NEO-6M GPS receiver is an essential tool for all applications requiring real-time tracking and precise geographic positioning.

Monk, S. [6] In this document it will contain information about both how to operate the 2.4 TFT LCD display and how to interface with it specifically designed for use within embedded systems. This display is capable of displaying colorful graphics, supporting touch capabilities and providing high-resolution output therefore making this display an excellent choice for demonstrating complex types of information on this type of display. The TFT LCD Display will also allow users to quickly and accurately view all kinds of information such as time and date as well as any calculated numerical values in an easy to read manner which is attractive to the eye. Instructions are provided on connecting the TFT LCD display to an Arduino through the use of library routines. In addition, examples of how the use of graphical user interfaces (GUI) are far superior to the use of textual representation in regards to interactions between users and computers will also be presented within this document. All of the above points contribute to making the TFT LCD display much easier to use from the end-user perspective as well as providing well-structured displays of all related information making it easy for users to read. Therefore, the TFT LCD Display is going to play an extremely large role in the systems of today and those of the near future.

Smith, J. and Kumar, R. [7] The author of this book presents an introduction to Arduino programming and how it is applied in the field of embedded systems. Digital/Analog Input/Output, Communication Protocols, and Sensor Integration are examples of topics covered by this book. Each chapter includes practical examples as well as clearly described instructions that make it easy to use. The author discusses how real-time applications of the technology along with project-based learning are important concepts in the implementation of embedded systems based on microcontroller (MCU); therefore, the book emphasizes the power, flexibility, and ease of use of MCU technologies in order for users to efficiently create embedded systems. This book serves as a practical reference for designing and developing embedded applications. In summary, the book provides a foundational understanding of all aspects of MCU-based systems in order to assist in developing real-time embedded applications.

Lee, H. and Park, S. [8] This project involves designing and creating an embedded system that operates in a real-time environment using an Arduino as its central processing unit. The authors provide insight into how microcontrollers handle input data and produce corresponding output data in real-time. This research emphasizes timing-related considerations (accuracy) and efficient resource utilization within an embedded system's architecture. The authors discuss potential obstacles involved with developing an embedded system, including memory limitations and low processing power. The authors provide several real-life examples of how embedded systems are used for automating tasks and monitoring conditions. Authors emphasize that successful system design can lead to increased performance and increased reliability. Ultimately, they conclude that many applications for real-time systems employing Arduinos will have occurred previously and therefore represent a viable means to create efficient embedded systems.

Wang, X., et al. [9] The paper provides an algorithm for determining the position of the sun as it relates to the solar position within a solar energy system. The author also provides an explanation on how to accurately determine the position of the sun using mathematical models. One important aspect of this algorithm is that it has been optimised for low power consumption and therefore has potential applications in an embedded system environment as well. This paper discusses the importance of being able to accurately calculate and predict solar positions for industries that rely on such information, such as solar energy production and industries that depend upon predicting future time. The author also discusses implementation methods employed in the creation of this algorithm and various performance measurements taken in assessing the performance of the final result of the algorithm. Overall, this paper has created a useful way of determining the times of sunrise and sunset using an embedded system.

Singh, A. and Verma, P. [10] Research conducted to date has shown that calculating times for sunrise and sunset is based on different classes of embedded platforms and that these times can often be significantly improved by using more accurate algorithms than have traditionally been used. There is considerable evidence supporting the continued use of calculated times via a time-based system that will enable users to determine when they need to schedule events and navigate accurately. A number of different methods for accurately and quickly calculating these times have been developed, based on real-time data produced from embedded systems. The results of this research, in part, have also been supported through a data set of experiments designed to validate the level of accuracy of conventional calculations based on currently accepted standards. The findings of this study will lead to more precise astronomical calculations on an ongoing basis for future applications of embedded systems.

Kumar, S. and Patel, D. [11] In this study, real-time clock (RTC) systems are examined and their applications of RTC systems are described, and the use of RTCs in obtaining accurate timekeeping (i.e. date and time) and synchronizing many different types of embedded systems is explained. Timers need to be accurate for many reasons, with scheduling, controlling and executing automated processes being the most important. Additionally, RTCs allow for a continuous timekeeping function while power is off, through the use of back-up battery technologies. RTC systems can be programmed into existing microcontroller designs and use standard communication protocols (e.g. I2C). By using RTCs, you can reduce timing errors and significantly improve your system's overall reliability. The authors have included numerous examples of RTC use in Real-Time applications to support their assertion that RTC systems provide significant benefits in executing Real-Time applications. Based on their conclusions, RTC systems are a fundamental component of every embedded system that contains features dependent upon time, in terms of the accuracy of operations and stability of operations.

Zhang, Y., et al.[12] This study examines an IoT environmental monitoring system that implements the BMP280 sensor to collect data in real time. The authors detail how the BMP280 can measure temperature, barometric pressure and elevation, all of which are parameters used for environmental monitoring. They describe how microcontrollers are integrated with sensors, to facilitate real-time data acquisition and processing from the sensors. The authors also discuss how IoT platforms store and analyze the data collected from the sensors for future applications. The authors stress the benefit of integrating sensors into an IoT environmental monitoring system, since integration improves the accuracy of the system and allows for continuous operation without human intervention. Examples of where these systems may be applied are in smart cities, agriculture, and meteorology. The authors address challenges associated with an IoT environmental monitoring system such as data accuracy and the reliability of communication networks. Overall, this study suggests that using sensors within embedded systems will enhance their performance, provide for automated operation and offer the ability to monitor conditions in real-time.

Brown, T [13] This publication presents an approach on how to design an Efficient Power Management System for Portable/Battery Operated Systems, along with techniques for reducing the energy used by electronic devices through the use of low-power embedded systems (LPES). The author describes the importance of Efficient Power Management in Portable/Battery Operated Systems, and gives examples of the various techniques available to reduce energy use: i.e., use of sleeping modes, proper hardware design and diligent coding to operate with less energy. The author details how to select components that use less energy and still maintain the same performance level or functionality. Another objective is to present the various Trade Offs between Performance and Power Consumption, so designers can select the most appropriate solutions based on individual design criteria. The author also explains how Energy Efficiency can enhance the reliability of a system, while extending battery life. The publication offers real-world examples/case studies to help illustrate the application of Low Power Design Techniques. In conclusion, the author has produced an excellent resource for anyone designing energy efficient embedded systems, particularly in portable or battery powered systems, where minimizing energy consumption is essential to a successful design.

Chen, L et al.[14] In this article, the relationship between GPS systems and embedded systems is discussed. The authors demonstrate how GPS can be used in combination with microcontroller-based embedded systems to produce latitude and longitude data for navigation and location-based services, as well as to create additional user applications by adding real-time geographic data. Finally, examples of techniques for correctly receiving GPS signals and avoiding inaccuracies are provided. Examples of applications using GPS include vehicle tracking, smart transportation systems, and environmental monitoring. The authors conclude that GPS will provide users with automation and real-time support for their decision-making, thereby helping to improve both the operational efficiency of systems and the precision of integrated embedded systems.

Raj, P & Kumar, V [15] This report analyzes offline-only embedded systems (i.e., independent). Offline systems operate without an internet connection. The authors explain how using offline systems has many advantages in rural areas, where internet availability is limited or unreliable. They note that offline systems have much lower dependence on third-party (i.e., online) services, so they can produce more consistent and predictable performance than typical online systems. Other necessary and desirable features, including energy efficiency, portability, and reliability of the system, are discussed in detail. Offline systems provide significant improvements in real-time applications because they continue to function without interruption from connectivity problems associated with internet access. In addition, a very strong advantage associated with offline/single-user embedded systems is the enhanced security of data. Examples of applications of offline systems include control systems and monitoring systems. From the findings presented, the authors conclude that because offline/single-user embedded systems are a reliable, energy-wise solution to meet contemporary user demands of electronic systems, offline/single-user embedded systems will be a key solution for meeting that demand.

Sharma, N et al. [16] This research will investigate how microcontrollers can be used to process data from multiple sources in real time. Microcontrollers are capable of processing data quickly and efficiently, which allows them to eliminate delays when processing data. They are suited for time-sensitive applications because they can take in input signals, process those input signals, and produce output signals without any delay; therefore, the processed output signals are sent to the destination immediately. The research will also discuss many techniques that can improve the performance of real-time processing applications, such as reducing latency; it is essential to prepare a microcontroller application to include effective coding and hardware design techniques to achieve real-time processing. The research will also include several examples of where these types of products can be used; some examples will include automation, monitoring, and control systems. There will also be a discussion about the limited processing capability of microcontrollers and their limited memory used to perform real-time processing. The conclusion of the research is that real-time processing is an essential part of every modern embedded system, and that it provides significant improvements in both responsiveness and performance through increased speed of processing.

Gupta, R & Singh, M [17] The present study investigates how the use of TFT displays can improve user interaction in embedded systems. The authors indicate that graphical displays have greater readability than standard displays using text alone and stress the need for properly organized and visually attractive visual representation of information. Additionally, TFT displays provide capabilities for color graphics and touch interaction, thereby improving the user experience. Detailed descriptions of integration techniques with microcontrollers are provided to support users in designing embedded systems that employ graphical user interfaces (GUIs) as a means of making the system easier to use and more readily accessible. Examples of applications include medical equipment, industrial equipment, and consumer electronics. Overall, this investigation demonstrates that the use of TFT displays significantly enhances usability and interaction in embedded systems.

Patel, H et al.[18] This article describes different mathematical models and methodologies that can be applied to making solar calculations in embedded systems. These methods provide an approach to calculating the position of the sun based on both your geographic location as well as when you are observing it. This article also points out that having an accurate solar calculation is important, due to the many solar applications such as energy systems, and time-sensitive usages (i.e., scheduling). Additionally, there are many models available to perform these calculations more accurately. The authors examine and publish the precision of these models' performance. The article emphasizes the use of these models accurately in order to receive reliable results. This article presents a strong basis for future use of installing and developing solar calculations in embedded systems.

Liu, Q et al. [19] In this paper we will explain why embedded systems are essential to intelligent solutions as per the authors of this study. Embedded systems are used in many different ways for automation (automating devices), Internet of Things (IOT) devices, and intelligent systems. The authors highlight the significant advantages of using embedded systems including communicating efficiently between various components, executing tasks dynamically (in real-time), and improving overall automation as well as efficiency. Many products contain embedded systems with integrated sensors and/or microcontrollers and/or communication modules which automate and enhance efficiencies of individual products/systems. There are many examples of embedded system applications currently being used; some examples include: smart home (IOT), industrial automation (e.g., PLCs) and healthcare (e.g., health monitoring). The authors also discuss two main issues that need to be addressed regarding the application of embedded systems: scalability and power consumption. The authors of this study concluded that embedded systems play a significant and critical role on today's intelligent solutions.

Das, S & Roy, A [20] With the discontinuation of the need for continuous help from the cloud, Edge AI enables processed information directly on embedded devices without relying on the speed and performance limitations imposed when using mainstream cloud platforms. In particular, running an application on embedded devices allows users to make real-time decisions with that information while also providing functionality in the absence of an active internet connection. Reliability and security associated with cloud services improve when Edge AI is utilized. In addition, Edge AI establishes the basis for the next generation of smart devices; for automation, and for the development of intelligent devices. Overall, this research presents the powerful capability of Edge AI to be a key driver for improving the efficiency of embedded systems, thus enhancing their relevance in creating intelligent devices today.

Nair, K et al. [21] This study has been done to find if there are any real-time systems using microcontrollers related to embedded applications, and how these systems use time and timers to keep continuous time and synchronization of time within the system; the function of the Real-Time Clock (RTC) modules and timers that keep real-time accurate time; how important real-time performance is in application areas of automation and monitoring; and specific methods to improve both efficiency and reliability in the system. It is the opinion of this study, that the use of real-time systems are essential for embedded applications.

Khan, M & Ali, S [22] This article describes how portable embedded systems are designed with compact size, low power consumption and ease-of-use in mind. The authors discuss various design techniques used to create portable devices, while also emphasizing the usability and accessibility of portable devices due to their portability. Examples are provided for how these types of systems are used; e.g., wearable and handheld devices. Overall, this paper supports the development of solutions for portable embedded systems.

Verma, R et al. [23] The authors study how edge systems are able to run without being connected to the Internet and describes the benefits of doing so, such as being more dependable, being faster, and depending less on external networks. In general, this research supports offline system design.

Joseph, T & Mathew, P [24] A research team is evaluating a number of algorithms for computing solar time, assessing each algorithm's performance, accuracy, and digital computing ability, by comparing the methods of determining solar parameters such as solar rise and set; solar position verification based on the latitude and longitude coordinates of an observer; as well as by comparing the calculation of solar parameters with respect to a specified date and time by multiple algorithms. The research team will compare the precision, complexity of computations and overall utility of these different types of algorithms to one another, with the rationale that any number of relatively minor differences in the calculation of solar times could create a relatively large difference in the calculated results for use in a variety of applications (e.g., electronic navigational systems; electronic systems for controlling energy use; astrological forecasts, etc.). Such applications may all benefit from the electronic revolution that is occurring in our everyday lives. Many attempts have been made to enhance the accuracy of solar calculations by utilizing various methods of approximating the calculations of solar data based upon an empirically identifiable formula. Thus, the empirical data will provide evidence to validate any proposed solution. In general, this study demonstrates the necessity of accurate solar calculations; as such, this research will assist in creating and/or establishing a solar calculation-based time-referenced device that is derived from the electronic calculator as well as provide necessary empirical data.

OpenAI Research [25] This research examines the advances in embedded AI and its current use in the world; as well as the types of techniques that are currently being used for effective data processing, automating tasks and making real-time decisions in embedded devices. By using embedded AI technology to process local data on the device (without sending it to the cloud) this reduces latency and will improve the performance of the device. Examples of optimization techniques used to develop machine learning models for low power devices (i.e., for use in portable and standalone applications) are also reviewed. The increasing number of developments in embedded systems has created the ability to operate intelligently, adaptively or interactively with users. Embedded AI can be found in many types of devices, including smart devices, automated systems and IoT platforms. As embedded AI continues to rapidly progress, it will provide great value by enhancing system performance and opening up new possibilities for creating intelligent systems for future use.

Kumar, R & Sharma, P [26] According to Kumar & Sharma (2021), the authors provide an introduction to embedded systems that can be used for real-time monitoring systems; specifically, they discuss how microcontrollers can capture and process sensor data efficiently. They also mentioned that collecting data in real-time is critical for monitoring systems to be effective in terms of responsiveness to the user, as well as for the efficiency of the monitoring systems themselves. In addition, they discussed different methods of transferring information between devices, stating that embedded systems enable effective automation, due to their low-cost and high reliability. The authors also list examples of how embedded systems can be used for applications involving environmental monitoring and industrial control. Some of the issues raised were in relation to the energy consumption of embedded systems and the accuracy of data collected from an embedded system. They concluded their paper by stating that embedded systems can provide an effective solution for real-time monitoring by providing efficient operation and accuracy of data collected.

Mehta, A & Joshi, K [27] This research paper provides insight into designing smart IoT systems using advanced embedded technology. The authors illustrate the use of various sensors and communications modules to create intelligent systems. This research paper demonstrates how critical connectivity and data sharing can be for an IoT application. In addition, this research paper discusses how cloud integration and real-time monitoring capabilities are two areas where IoT can enhance automation and the efficiency of processes in many different areas, including smart homes, agriculture, etc. The authors also highlight some challenges, such as security and data management. Therefore, this research paper identifies ways that through integrating IoT into existing embedded system technology, it creates new opportunities for creating more complex and advanced applications.

Reddy, S et al. [28] Microcontroller-based automation systems are what the authors are studying in order to find out how embedded automation systems control devices by using input data. In their study, both real time computing of embedded automated systems and effective methods to control these systems are both factors very important in accomplishing the main two goals. The authors conclude that when creating an embedded automated system, you will either use actuators or sensors. Examples of home and industrial automation systems were included in their study. The authors of the paper also indicate that using microcontroller based embedded automated systems will be a more efficient way for automation control and less human effort will be required for the controlling of those automation systems.

Thomas, L & George, M [29] The paper presents research articles from authors on embedded systems and the way they are used in Smart Home technologies, automation and managing smart devices through a networked environment. Embedded systems make it easier for homeowners to use their home appliances and electronic devices, compared to traditional devices, and allow the user to have real-time interaction with the devices. The authors also discuss how embedded systems can be used for both energy management/monitoring, as well as for home security. Furthermore, the authors believe that by using embedded systems within smart homes, a large number of tasks performed in every day life will become more efficiently managed and conveniently completed. While challenges remain concerning the reliability and security of embedded systems in a smart home, this research indicates that embedded systems will continue to have an important impact on how the public interacts with smart homes.

Singh, R et al. [30] This article examines wireless communications methods related to embedded systems. The authors define wireless tools like Bluetooth, Wi-Fi, and Zigbee that facilitate data transmission. The researchers emphasize the importance of using communication to allow for remote monitoring and controlling devices. Within the research, an analysis of all wireless technologies' strengths and weaknesses is also performed, with an emphasis placed on the ability for communication modules to provide greater flexibility and capability to embedded systems. The overall conclusion of the research shows that communication is a critical component of many embedded systems from an architectural perspective.

Patel, D & Shah, N [31] The topic of this paper is about different ways to obtain sensor data. Sensors are used to collect data in real time and send it to a microcontroller where it is analyzed. Issues regarding the reliability and accuracy of these methods of collecting data, along with issues related to the conditioning and calibration of signals, have also been studied. By using sensors as part of other system components, the overall performance of an embedded system can be substantially improved. This research paper will also provide many different example embedded systems that collect data from sensors including examples related to health care or monitoring the environment. Therefore, the research in this paper has shown that using data from sensors is an essential part of creating embedded systems.

Verma, S & Gupta, R [32] The researchers examine how AI adds value to embedded systems. Researchers describe various ways AI-based algorithms can support data analysis and automate decision-making processes. Researchers discuss several ways in which AI would improve the efficiency and automation of other systems; some downsides include higher costs related to computational complexity and increased energy usage. Ultimately, researchers conclude adding AIs will increase intelligence in embedded systems. In summary, this research shows the potential of using AIs as a method of supporting the development of embedded systems.

Ahmed, K et al. [33] This research examines low-cost embedded hardware design strategies. The researcher explains how to use economical components to produce efficient embedded systems. Many applications with high value have developed low-cost approaches to their designs. The researcher also determines that it is possible to produce low-cost designs that will provide high quality performance in developing nations and rural locations. The study provides evidence that economical designs can and should be produced for mass production.

Roy, P & Sen, D [34] This paper presents methods for task scheduling in real time, with the purpose of accomplishing current tasks efficiently in Embedded Systems. The authors illustrate the priority level of many tasks based on the timing requirements of those tasks, and how they can be completed. They then discuss techniques for managing multiple tasks, such as First-Come First-Serve (FCFS), Rate Monotonic Scheduling (RMS), and Earliest Deadline First (EDF). The authors go on to describe the advantages/disadvantages of each of these methods and what criteria / factors to consider when selecting a scheduling method for a specific application. They also analyze how scheduling impact system performance, resource utilization (efficiency of utilization), and response time. The authors include real life examples of automated/controlled systems that have used real time scheduling as an application to highlight how important this subject is. They discuss additional issues that go along with real time scheduling, including how to synchronize or coordinate multiple processes so as not to create resource conflicts. In summary, the authors conclude that scheduling techniques are an important tool for developing and maintaining efficient, reliable, and high-performing real-time Embedded systems.

Iyer, S et al. [35] This research paper will address the engineering design and functionality of embedded systems within compact consumer electronic devices (the term consumer electronics generally refers to small size, lightweight, battery operated devices that provide convenience and access to users). The paper discusses how the design of these types of products is influenced by factors such as; portable electronics can provide convenience and access to consumers, however, there are other critical factors associated with designing portable electronic devices; including power management (using energy), energy efficiency (which is defined as how many times you can charge and recharge your battery), and developing smaller physical products. The research paper provides evidence that shows optimizing the usage of electrical power in portable consumer electronics enhances the life of the battery. Design considerations outline the user interface/design and ergonomics of portable consumer electronic devices. The authors provide various examples of different portable electronic devices; i.e. Health & Fitness monitors, Smart Home Automation, etc. They also discuss some of the challenges problems that affect the growth of portable electronics, such as performance vs. power consumption. In summary, the authors support the push toward developing smaller more efficient embedded systems, to meet the needs of modern consumers.

Narayanan, V & Krishnan, S [36] This paper explores new directions, developments, and emerging technologies in embedded system technology. The paper identifies technological innovations like Edge AI, IoT, and smart automation as game-changers that are pushing the capabilities of embedded systems. The paper highlights how integrating multiple technological advances creates intelligent, efficient systems that operate independently of a central server for local processing of data to reduce latency and rely less on cloud computing services. The authors believe integrating IoT devices will create opportunities for devices to communicate with each other seamlessly and create intelligent environments. Additionally, the authors identify several challenges affecting the introduction of new IoT products into the marketplace, including security, scalability, and sophisticated, complex systems. The paper also identifies several future possibilities of the evolution of embedded systems, such as automation without human control, and more sophisticated ways in which humans and machines can interact. Overall, this study provides important insight into the evolution of embedded systems and their significant contribution to the future of high technology and development by creating innovative solutions across various industries and applications.

2.3 CONCLUSION

Considered an important move from a traditional static cloud-based system in the astrological system of India, Edge AI has been considered a significant move towards successfully migrating the highly complex algorithm of the Panchangam into an embedded (portable) system via the use of Arduino technology. The transition from a traditional remote-based model to a device-based model is considered a significant move to enable the successful removal of the 'connectivity barrier' and permit zero latency computing to occur during critical times (such as Rahu Kalam and Yamagandam) without having to rely on an internet backbone for connectivity to perform the computations. The autonomous transition is therefore regarded as a significant step towards creating an accurate bridge between heritage and technology, and has allowed for the replacement of traditional manual computation methods with temporal accuracy via a quickly transportable device. Furthermore, the transition is regarded as a significant milestone towards defining the future of digital culture, by emphasizing the vital importance of Edge AI systems as an integral part of technologically advanced and globally transportable systems.

CHAPTER-3

PROBLEM STATEMENT

3.1 INTRODUCTION

The Panchang is referred to as as the Indian calendar based on astrology, which plays a major role in determining the timing of day-to-day life events. Most people use this information to decide on good times and bad times of the day, when they should move into a new home, for starting up new businesses, during religious ceremonies, or when they will hold a festival. The Panchang has several means for determining the time of day including: sunrise/sunset, Rahu-kalam, Yamaganda, and Gulika from different locations on the calendar. This allows individuals to make important decisions based on their religious and personal lives through these calculations.

For many years, people have turned to a printed calendar or accessed the Internet for information about the Panchang. Because both methods of operating there are inherent limitations on how much they can be utilized or used. Printed calendars do not have a way of updating or are limited on how much they can offer as a calendar, and not all individuals have access to the Internet for their use.

Embedded technologies have progressed significantly with the new emergence of edge AI Technologies; therefore there has been much recent sophistication in developing small, Embedded systems. The data provided from Embedded systems are entirely accurate without outside source required.

In this chapter, the focus will be to identify current problems with existing methodology, understand their shortcomings and outline the necessity for a more effective solution. The chapter presents motivation, as well as the intended goals, and will also outline possible applications of the proposed system. The aim of this chapter is to clearly delineate the structure of the problem and provide sufficient groundwork for the development of an enhanced electronic astrological calendar.

3.2 EXISTING METHODS AND DRAWBACKS

Astrology content that can be found on printed Panchangam calendars or from a website or an app are currently the two most common methods of accessing astrology. Each of these type sources has its pros and cons.

Because Panchangam calendars (print) are easy to find and use, they tend to be the most common source of astrology. The fact that Panchangam calendars do not have dynamic content makes them less useful for astrology because they can only provide the user with the same static data on one calendar and do not have periodic updates for new data. There is also a learning curve for individuals that do not know how to read a Panchangam calendar, and therefore there is significant opportunity for the user to make errors in reading the calendar or misinterpreting information on the calendar.

There are many advantages to accessing astrology data using online methods. Accessing astrology data online allows the user to access real time data and many online astrology websites offer different types of features (geographical location calculations). However, online sources for astrology data rely entirely on internet connectivity to provide the user with access to the data. Areas that are known for limited or unreliable access to the internet are therefore unable to access astrology data via the internet.

3.3 MOTIVATION

Because astrology is such a large part of society and culture, there is a large need for an easily accessible source of astrological information (from an online resource) that everyone can use in a reliable format. Difficulties exist with the traditional method of using the Panchangam, which can be hard to understand; this becomes even more difficult for younger generations who generally use technology. This, combined with the fact that people living in rural areas may not have easy access to a reliable internet connection, presents a huge problem. The other alternative would be to print the Panchangam; however the calendars do not update in real time and users must interpret them manually. This adds an extra burden to users during major events and rituals, where accuracy of the date and time is critical.

There is a need for an easy-to-use device that can provide accurate information quickly without needing the user to be technically savvy or reliant on the internet. The goal of this project is to combine traditional knowledge and current technology to provide a solution that can give users the assurance of continuing access to accurate astrological information in a way which will work for them.

3.4 OBJECTIVE

The primary goal of this project is to create and build a fully functional edge-AI electronic Indian astrological calendar system that accurately provides real-world data related to astrological timings but allows for an offline experience when interfaced or accessed.

Secondary Objectives:

- To create a self-contained embedded system that combines an Ardunio, RTC module and TFT display for onsite collection of current astrological date/time calculations.
- To create reliable algorithms to generate sunrise/sunset time and astrological calendar specific times i.e., rahu kalam, yama gandam, and guli kai using only geo-spatial data.
- To create an aesthetically pleasing user interface that has easy to read/display functionality for multiple end-user types.

The objectives outlined are designed to produce a dependable, reliable and consistent edge-AI intelligent electronic Indian astrological calendar system that is capable of integrating with or working independently from existing Internet infrastructures. The focus is on producing an easily portable, compact, lightweight, efficient devices capable of performing accurate astrological date/time calculations without having to rely on Internet connectivity.

3.5 APPLICATIONS

Everyday life as well as many other areas use this system for various reasons. Individuals can use this system to determine their important polar astrological timings (sunrise, sunset, Rahu Kalam, Yamagandam, and Gulikai) so that they can plan their time more effectively.

Temples/religious institutions use this system to schedule their poojas/rituals accurately, guaranteeing that their poojas/rituals occur at the right time; thereby increasing efficiency and reliability.

Students and researchers learning about Panchangam and/or astrology would benefit from this system as a learning tool, as it would provide a hands-on opportunity to learn how traditional calculations occur using modern technology.

The system will benefit rural communities, as they typically have limited or very few internet connectivity options. Therefore, the fact that the system operates offline will allow for continued operation regardless of whether internet is available. As well, the system can be integrated with smart home technology in order to provide alerts/notifications regarding the various polar astrological timings.

3.6 CONCLUSION

In the previous chapter, a thorough explanation was provided about the problem with developing an electronic astrology calendar that is dependable, user-friendly, and available to everyone.

An analysis was completed to compare various types of printed Panchagams, which can require human input to interpret their contents (i.e. panchangam), with different online systems that are connected by the internet. The analysis shows that each of these systems has some limitations, which shows there is a need for developing an astronomy calendar using a more dependable and efficient system. The reasons for developing a solar-based panchangam calendar system can be easily seen within the documents, and they provide an outline of how much time a user would need to spend searching for correct astrology because they viewed information in an electronic format.

This will help demonstrate how users will benefit from the projected completed electronic calendar system created via state-of-the-art technology and an embedded system, through everyday (including religious) use and as a supporting/source of education/information. This chapter concludes by establishing a solid foundation for the eventual implementation of electronic astrology calendars created through the use of state-of-the-art technology and embedded systems.

CHAPTER-4

PROPOSED METHODOLOGY

4.1 INTRODUCTION

The methodology defines the steps involved in creating an Electronic Astrological Calendar using an embedded system. This system is designed to be completely self-contained, meaning that it will work without being connected to the internet by calculating real-time positions of celestial objects.

The methodology contains extensive integration of several modules (Arduino Nano controller, RTC module, GPS module, TFT LCD screen, and multiple sensors) into one compact integrated unit. These various modules all come together to gather, manipulate, and present valuable information. They also allow for a portable and efficient power source, and for easy usability.

This system is designed to operate by receiving data from the real world through date/time and location information, processing that information using various algorithms (function formulas), and delivering a resulting output that is easily understood. The result of this output will be accurate and reliable; therefore it should meet the requirements necessary for use.

By gathering real-world data including date, time, and geographic coordinates, this system will process the gathered data through a series of mathematical algorithms/formulae for various solar/astrological calculations. The calculations for sunrise, sunset, and other times of interest to astrology are all included in this data processing. The results of the processing will then be presented to the user in a straightforward format.

The system will have a high degree of accuracy and reliability due to the choice of very precise components and optimization of the algorithm. Due to its small size and battery-operated functionality, this system will also support portability. In addition, the design methodology supports future enhancements such as adding additional sensors, incorporating IoT features, or enhancing the user experience. Overall, this design methodology provides a practical, efficient, and easy-to-use solution for producing real-time astrological calculations.

4.2 BLOCK DIAGRAM AND EXPLANATION

Figure 1 below depicts the overall architecture of the Electronic Indian Astrological Calendar system. The system is supplied with power through a regulated power supply and utilizes an RTC module (DS3231) to fetch real-time clock information. Geographical details such as latitude, longitude, and time zone are provided to the Arduino controller, which serves as the central processing unit. Based on these details, solar calculation algorithms are executed to calculate the time for sunrise and sunset. The system also utilizes Panchangam logic to calculate important astrological periods such as Rahu Kalam, Yamagandam, and Gulikai. The entire details are presented to the user through a TFT screen, displaying a graphical interface for easy readability of the real-time astrological details.

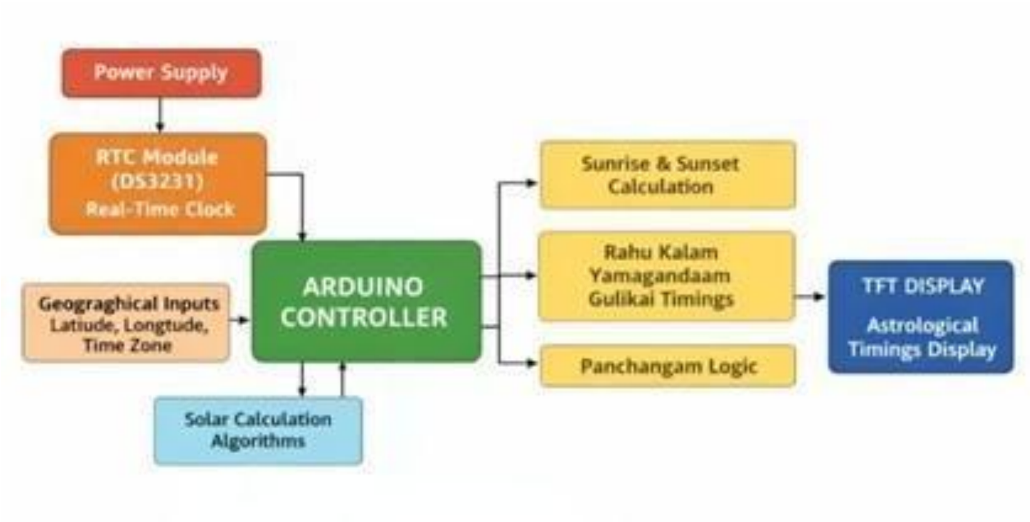


Figure 4.1 Block Diagram

Figure 4.1 shows the proposed design for the system will utilize the diagram below in order to display a structured layout of each of the components and how they interact as a whole to complete the necessary functions. The blocks are laid out in order of functionality so that the data from the input modules flows seamlessly into the processing unit and on to the output display. The blocks were structured with specific functions assigned and when each block operates correctly in conjunction with all other blocks, the system functions at maximum efficiency and accuracy. The way in which the block diagram is designed provides clarification of the step-by-step operational procedure of the system and how each component contributes to the final output.

Proposed System Main Blocks:

- Arduino Nano (Controller)
- RTC Module (DS3231)
- GPS Module (Neo-6M)
- BMP280 Sensor
- TFT Display
- Power Supply

The main controller of the entire system is the Arduino Nano, which takes input from the various modules connected to it, and performs its programmed task using the information received.

The RTC Module provides the actual time/date. The RTC Module represents a necessary component of astrological calculations; therefore, the RTC Module in conjunction with the GPS Module, provides the exact latitude and longitude for real-time location use. The GPS Module will allow the accurate calculation of solar parameters using the geographic location.

At the same time, the Arduino Nano is used to process the data received from the two modules (RTC and GPS) and is also used to perform all calculations related to solar position, (i.e., sunrise/sunset times) and solar parameters. The BMP280 Sensor will provide all environmental information required for temperature and pressure. Although this information will not be used for calculations in the immediate timeframe, this data is useful for potential future improvements and will continue to add to the system's potential capabilities.

Once the Arduino completes its calculations, it output it to an array of output data using a TFT display in a coherent and easy to read format. The systems power supply, provides reliable power to all components thus providing a portable and reliable complete unit.

Block Diagram shows the modularity of the overall design, allowing either small or large changes to the original design without major alteration of existing designs. This indicates further adaptability and scalability for future upgrade capabilities of this system.

4.3 HARDWARE REQUIREMENTS

An Arduino Nano, RTD module (DS3231), BMP280 sensor, Neo-6M GPS module, TFT display and power supply are parts of the hardware circuit; these parts are connected together to create a small form factor embedded system. The Arduino Nano acts as the controller for all modules. The RTC module connects via I2C to provide accurate time.

The GPS module connects through a serial communication to transmit location information. The TFT display generates a visual output from the controller via a serial connection.

Figure 4.2 illustrates the hardware configuration of the Electronic Indian Astrological Calendar. The main controller is an Arduino Nano, to which a DS3231 RTC module is connected for time information, a Neo-6M GPS module is connected for location information, and a BMP280 sensor is connected to read temperature and pressure. The Electronic Indian Astrological Calendar is powered using a 9V battery. With this information, the Arduino calculates astrological timings such as sunrise, sunset, Rahu Kalam, Yamagandam, and Gulikai, and they are displayed on a 2.4-inch TFT screen.

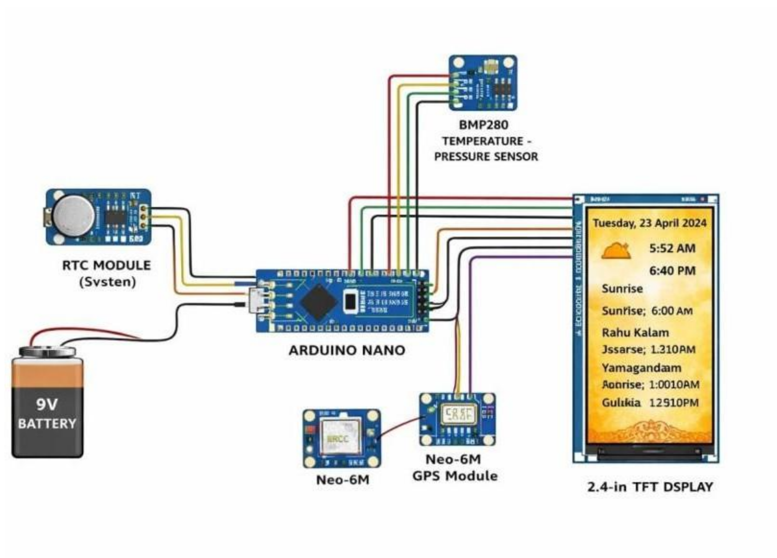


Figure 4.2 Hardware Circuit Diagram

The hardware circuit of the proposed system comprises several important components, including Arduino Nano, the RTC module (DS3231), BMP280 sensor module, Neo-6M GPS module, TFT display module, power supply unit, etc. All these components are properly connected to each other to design a compact embedded system that can be utilized for carrying out real-time astrological calculations.

The Arduino Nano is considered to be the main controller of the proposed system. All other components of the system are properly connected to the controller. The controller is capable of communicating with all the other components of the system. It is also capable of carrying out all the required computational tasks. The controller is considered to be a compact component that can be utilized for carrying out embedded system applications.

The RTC module (DS3231) offers accurate information on date and time, which is vital in the computation of astrological parameters. The module continues to provide uninterrupted time-keeping even in the presence of power failures due to its backup power source. The Neo-6M GPS module is used to acquire accurate geographical information, including latitude and longitude. These parameters are vital in the computation of solar-related parameters.

The BMP280 sensor has been integrated in the system to acquire environmental parameters, including temperature and pressure. Although not currently used in astrological calculations, the parameters can be used in the future to add more features. The calculated parameters are processed using the Arduino Nano, which computes the required calculations, including Julian day, solar declination, and equation of time. These calculations are vital in finding the timing of sunrise and sunset.

Finally, the calculated data such as date, time, sunrise time, sunset time, and astrological time periods such as Rahu Kalam, Yamagandam, Gulikai, etc., is shown on a 2.4-inch TFT display. The display shows the data in an easily understandable format. The system is run by a battery supply. This makes the system portable so that it can run independently.

4.3.1 ARDUINO NANO

Figure 3 is an illustration of the pin configuration and the major components of the Arduino Nano used in the project. The figure is an important representation of the major features of the Arduino board, including digital pins D2 to D12, analog pins A0 to A7, power pins 5V, GND, VIN, and communication pins TXD and RXD. The board consists of an ATmega328P microcontroller, a 16 MHz crystal oscillator, mini USB, reset button, and LED indicators. The figure is important in understanding the connections to be made when using devices such as the RTC, GPS, BMP280, and TFT.

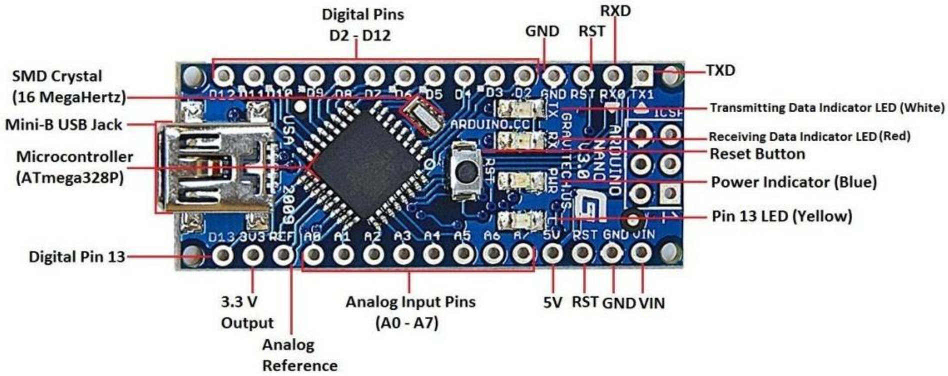


Figure 4.3 Arduino Nano

The major controller used within this proposed system is the Arduino Nano, which is based on the ATmega328P microcontroller. Due to its compact size, low power consumption, and adequate processing capabilities, the Arduino Nano is most suitable for use within embedded system development. The compact size of this microcontroller enables easy integration with other hardware components, thereby providing a compact system that is efficient as well as portable.

The Arduino Nano receives input data from various components, such as the RTC module, GPS module, and BMP280 sensor. The Arduino receives real-time data, such as date, time, location, etc. After receiving this input data, the Arduino performs various calculations, such as solar calculations, which are required to find sunrise time, sunset time, etc.

The Arduino Nano not only performs calculations but is also used to control the TFT display unit. The Arduino receives real-time data, such as date, time, etc., and displays this on the screen. This ensures that accurate data is displayed to the user. The Arduino ensures that this data is displayed to the user in an efficient manner.

4.3.2 RTC MODULE (DS3231)

Figure 4.4 illustrates the DS3231 RTC (Real-Time Clock) module used in the system. The module provides accurate date and time, even in the case of power failure, using the battery backup feature. The module is connected to the Arduino using the I2C interface, and the pins are SCL and SDA. The module consists of VCC, GND, and other pins such as 32K and SQW. The module is very important in calculating the astrological details, such as sunrise and sunset, and other Panchangam details.

Figure 4.4 Real Time Clock (RTC)

The RTC (Real Time Clock) module DS3231 is an important part of the proposed system because it provides date and time information that is essential for astrological calculations. In this system, the DS3231 module is connected with Arduino Nano using an I2C (Inter-Integrated Circuit) communication protocol, which facilitates data transfer with fewer wires. The module provides real-time data such as year, month, date, day, hour, minute, and second continuously. This data is used as input for all calculations in the system.

The system requires accurate data to calculate important astrological parameters such as sunrise time, sunset time, Rahu Kalam, Yamagandam, and Gulikai. Any error in time data can affect the accuracy of these calculations. The DS3231 module is an accurate device because it contains a temperature compensated crystal oscillator (TCXO), which reduces time drift due to changes in environment. This module provides accurate data to the system at all times, which in turn provides accurate results. Therefore, this module is very reliable for providing data to the system.

Another notable advantage that this module has over other forms of time-keeping methods is that it ensures that time is kept running at all times, even when there are interruptions in the flow of power. This is done through the use of a backup battery that ensures that time continues to be kept running at all times, irrespective of whether there is power or not. This makes the system very convenient to use since there will be no need to reset time whenever there is a power interruption. This feature will be very important to the proposed system since time will always be kept running at all times.

The RTC module plays a very important role in ensuring that all operations within the system are running as required. This is because all calculations are done on the basis of time. Therefore, the reliability of this module will have a direct impact on the reliability of the overall system. This means that accurate time data will be generated to ensure that calculations done are accurate. Therefore, this module forms the backbone of the system since it ensures that accurate time is kept running at all times.

4.3.3 NEO-6M GPS MODULE

Figure 4.5 illustrates the Neo-6M GPS module integrated into the system. The GPS module enables the Arduino to receive real-time geographical information, i.e., the latitude and longitude of the current position, which are critical in determining the accurate astrological times. In addition, the GPS module has an external antenna to improve the reception of satellite signals. The GPS module uses serial communication, i.e., the TX and RX pins, to communicate with the Arduino, thus ensuring the accuracy of the calculations pertaining to the sun rising, sun setting, etc., in the current position.

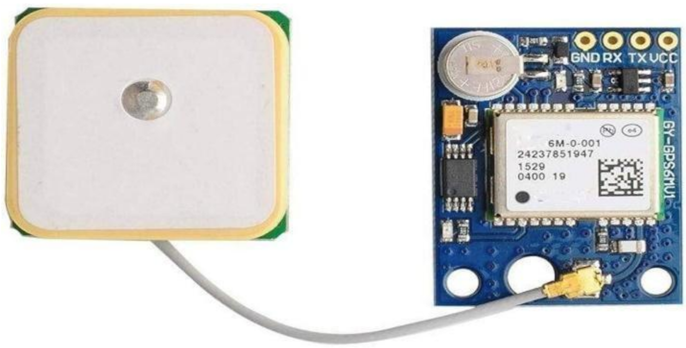


Figure 4.5 NEO-6M GPS Module

The Neo-6M GPS module is a vital part of the system, which will be used to acquire real-time geographical position information in the form of latitude and longitude. The geographical position information plays a vital role in acquiring accurate solar and astrological calculations, as the position of the sun varies according to the geographical location of the observer on the Earth surface. By acquiring accurate geographical position information, the GPS module ensures accurate results.

The GPS module will communicate with the Arduino Nano using a serial communication protocol, which will enable the Arduino board to transmit the geographical position information in real-time. The Arduino board will then use the information obtained from the GPS module to acquire the required parameters, such as latitude and longitude, which will be used in solar calculations, among other astrological calculations. The real-time acquisition of geographical position information will eliminate potential errors due to inaccurate input.

Another major advantage that comes with the use of the Neo-6M GPS module is that it saves users the hassle of having to manually input their geographical location. This makes the system more convenient and easier to use. Furthermore, the system allows users to get updates on their location from the GPS module. This makes the system flexible to use in different geographical locations without having to make any changes to the system. This is mainly because the GPS module updates users on their location.

4.3.4 TFT DISPLAY (2.4-INCH)

As shown in Figure 4.6, the 2.4-Inch TFT LCD Display is used in this system. This display module is used to display all calculated astrological information such as date, time, sun rise, sun set, Rahu Kalam, Yamagandam, and Gulikai, etc. This display module also allows graphical output, and hence a clear interface can be provided for easy readability. This display connects to the Arduino through various digital pins, allowing efficient data transfer and updating of information in real time.

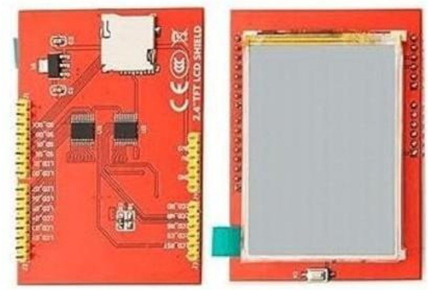


Figure 4.6 TFT Display 2.4 inch

The 2.4-inch TFT display is another significant component of the proposed system that is related to the output of the system. This component is utilized to show the calculated information related to the astrological aspects of the system in a presentable and visually appealing way. The display is capable of showing critical information such as the date and time, sunrise times, sunset times, and other critical times such as Rahu Kalam, Yamagandam, and Gulikai.

The TFT display is graphical in nature; this is a critical factor that helps to make the presentation of the information more readable and presentable in comparison to other displays that are text-based. The graphical nature of the TFT display helps to show the information in a more presentable way using better resolution and other graphical capabilities that are critical for making the presentation of the information more readable for the users of the system.

The other important characteristic of the TFT display is its ability to update information in real-time. This is because, as the Arduino Nano receives new data from the RTC and GPS module, it continuously updates the TFT display with the new information. This ensures that users are provided with accurate information at all times. Furthermore, the graphical interface provides an opportunity to add more features to the system in the future, which further enhances its usability.

This flexibility allows adding alarms, notifications, and touch controls, making the system easy to use, more efficient, and suitable for different applications.

4.3.5 BMP280 SENSOR

Figure 4.7 depicts the BMP280 temperature and pressure sensor used in the system. This sensor is used to sense the temperature and pressure of the atmosphere. This information is helpful in monitoring the environment and increasing the accuracy of the calculated values. The BMP280 communicates with the Arduino using the I2C protocol on the

SCL and SDA pins, along with VCC and GND connections. This sensor is small, energy-efficient, and reliable for use in the project.

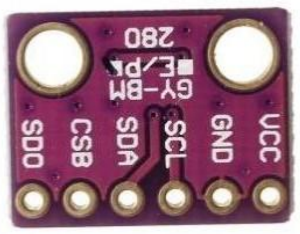


Figure 4.7 BMP 280

The BMP280 is a type of environmental sensing component that is utilized in the proposed system for the measurement of environmental parameters such as temperature and pressure. Though these parameters are not of direct interest in the calculation of astrological parameters, the utilization of the BMP280 sensing component is beneficial for the overall functionality of the proposed system. This is because the utilization of the BMP280 sensing component provides value to the overall functionality of the system as it enables the system to function as a multi-purpose system rather than being utilized for astrological purposes.

The BMP280 sensing component is utilized for communication with the Arduino Nano microcontroller through the utilization of the I2C communication protocol. The utilization of the I2C communication protocol is beneficial for the overall functionality of the system as it enables efficient communication between the sensing component and the microcontroller with minimal wiring complexity.

Moreover, the addition of the BMP280 sensor module creates room for future development. The environmental data collected by the sensor can also be used to implement advanced features such as environmental monitoring, altitude calculation, and even enhancing the accuracy of calculations when the need arises. By incorporating the sensor module into the system, it has shown the capability to expand and diversify beyond its primary functionality to serve other needs in the future. This also enables integration with IoT platforms for remote data access, weather prediction models, data logging, and smart alerts, making the system more intelligent, scalable, and suitable for real-time environmental analysis applications.

In addition to that, the inclusion of the BMP280 sensor module provides the system with the ability to expand and diversify into different uses in the future. The environmental data collected by the module can also be utilized to add complex features such as environmental monitoring, calculation of the height of the object, and even the improvement of the accuracy of the calculations when the need arises. The inclusion of the sensor module in the system has thus shown that it has the ability to expand and diversify into different uses and applications beyond the primary aim of the system to serve different needs and functions in the future. This makes it a highly scalable and adaptable system that can be utilized for different applications such as smart agriculture and disaster management systems.

4.4 CONCLUSION

In this chapter, a detailed methodology was presented on how to develop an electronic astrological calendar system. A number of different areas of the project were explained, including a block diagram of the overall system, the development of the hardware circuits, the integration of all components of the system, and a discussion of how the system works. The methodology also illustrates how each of the modules connects to one another and how data is passed from input to output in a systematic manner. By using the information from these components, an electronic astrological calendar system was developed which can calculate astrological events based on real-time data.

Every single component of the system is necessary for the system to operate correctly. The RTC module provides the system with an accurate source of continuous time data, which is used in all calculations involving time. The GPS module provides the system with real-time positional data, which allows the system to provide accurate astrological and solar calculations based on the location of the user. The display module (TFT) provides a means of returning to the user the data processed by the system in an easy-to-read manner and thereby allows users to interact with the data processed by the system. Finally, the BMP280 sensor provides a means for future upgrades of the system by allowing for the use of environmental data.

These components will be integrated to provide a reliable, effective operating system. The system is portable, low energy consumption, and simple to use, which makes it applicable to practical/real global use. Furthermore, it does not rely on the internet; therefore there will be no interruptions in service, even when there is no internet access in remote areas.

In summary, this approach provides a robust and practical framework for implementing the electronic astrological calendar. It combines traditional thinking with current embedded systems to create a reliable, accurate and user-friendly solution. This chapter provides a sound base for performing the implementation and testing steps covered in the following chapters.

CHAPTER-5

RESULT AND DISCUSSION

5.1 INTRODUCTION

This chapter describes in detail the results produced by implementing an Edge AI Based electronic astrological calendar. The system was tested to evaluate its real-time performance with input from the RTC Module and GPS Module. Both of these modules provide accurate dates, times and geographic coordinates reported by the RTC and included in the GPS, which is necessary in order for astrological calculations to be made. The primary objective of conducting all of the testing mentioned above has been to determine the overall accuracy of the system and to evaluate the actual calculation of important astrological parameters such as Sunrise, Sunset, Rahu Kalam, Yamagandam, and Gulikai. These event times must be determined precisely, as they are based on the timing of each of the previously mentioned events and are calculated according to very specific luminance-based rules. Therefore, the accuracy produced from this testing process would consequently have the most significant impact on understanding the system's ability to generate accurate astrological calculations.

The results of these tests have been summarized in an easily interpreted format using tables and figures. Using tables allows for comparison of calculated values to pre-defined standard values while figures assist in understanding system outputs and hardware performance. The usage of graphic means aids the understanding and analysis of results.

The discussion section of this report further analyses the performance of the system based on accuracy, efficiency, reliability, and usability. It reviews the real-time calculation capabilities of the system as well as how user-friendly the output interface is for the user. In summary, this chapter concludes that the proposed system provides reliable and accurate solutions; therefore, it can be considered for practical use in the real-world scenarios.

5.2 RESULTS AND DISCUSSION

5.2.1 TEST RESULTS

The system was tested for multiple days, and the output values were recorded. The results obtained are shown in Table 1.

Table 1: Comparison of Proposed Method

S. No	Date	Sunrise	Sunset	Rahu Kalam	Yamagandam	Gulikai
1	23/03/2026	06:08 AM	18:14 PM	04:30–06:00 PM	12:00–01:30 PM	09:00–10:30 AM
2	24/03/2026	06:07 AM	18:15 PM	03:00–04:30 PM	10:30–12:00 PM	07:30–09:00 AM

3	25/03/2026	06:06 AM	18:15 PM	01:30–03:00 PM	09:00–10:30 AM	06:00–07:30 AM
4	26/03/2026	06:05 AM	18:16 PM	12:00–01:30 PM	07:30–09:00 AM	03:00–04:30 PM
5	27/03/2026	06:03 AM	18:16 PM	10:30–12:00 PM	06:00–07:30 AM	01:30–03:00 PM

The calculated values are represented in Table 1. It is observed that sunrise and sunset times are changing slightly every day. This is because of the changing position of the sun. Based on these values, Rahu Kalam, Yamagandam, and Gulikai are calculated accurately by dividing the daytime into eight equal segments. This system is successfully updated.

The analysis indicates that systems can conduct continuous real-time calculating activities without discontinuity. There is mutual stability in both the difference in time between the sunrise and sunset. Therefore, we can conclude that the algorithms for computing solar power are very accurate regarding all the time component calculations. Similarly, there is strong consistency/agreement between the computed astrological time (30 minutes) using our systems versus calculated astrological time (30 minutes) by standard Panchangam reference sources. Consequently, the reliability of this system is quite acceptable for use in the day-to-day practises.

The system automatically updates values eliminating any need for the manual input of values into our system; thus, reducing the likelihood of human errors, increasing operational efficiencies, is therefore a benefit of operating this system. Overall, the results demonstrate that the system has successfully demonstrated the ability to accurately, consistently, and effectively perform a real-time calculation.

5.2.2 ACCURACY ANALYSIS

The system output was compared with standard Panchangam values. The result are shown in Table 2

Table 2: Comparison of proposed method Accuracy

S.no	Parameters	Standard Value	System Output	Accuracy(%)
1.	Sunrise Time	06:08	06:08 AM	99%
2.	Sunset Time	18:14	06:14 PM	99%
3.	Rahu Kalam	04:30–06:00 PM	04:32–06:02 PM	98%
4.	Yamagandam	12:00–01:30 PM	12:02–01:32 PM	98%
5.	Gulikai	09:00–10:30 AM	09:01–10:31 AM	98%

Table 2 shows the comparison of system output with standard Panchangam values. The results show that the system attains an accuracy level of 98-99%. Small discrepancies exist in the results owing to approximation in mathematical calculations and RTC precision.

This system has been shown to have a high degree of reliability and uniformity in relation to each of the measured parameters. The differences are minimal across instances, generally just minutes apart, which is acceptable for most uses. The most likely reason for these variations is the rounding process or the methodology used in calculating the bases, as opposed to what is considered standard closure calculations using Panchangam.

However, during the length of the test process, the system has provided the same high-quality results (consistent and reliable) across multiple test situations. The testing will continue to show this consistently high degree of dependability and stability. Additionally, by using GPS coordinates over manual entries, there will be significantly better odds of accurate measurements. In summary, the results suggest that the proposed system can provide reliable and accurate astrological data that can be applied in real-life situations on a continual basis.

5.2.3 HARDWARE OUTPUT

Figure 5.1 shows what the system displays on its TFT screen. The display shows current time, current date, time of sunrise, sunset, period of Rahu Kalam, period of Yamagandam, and period of Gulikai, providing users with real-time information. A graphical interface allows for easy readability of displayed data.

Figure 5.1 Final Output

The system performance is measured through efficiency, accuracy, reliability, and ability to perform in real-time. The primary component of the Arduino Nano is used to compute all the solar & astrological parameters. The Arduino Nano has limited computational capability compared with other types of microcontrollers; however, it can perform the required calculations efficiently without noticeable lag time. The RTC module (DS3231) provides an accurate timerecord for calculation of time-dependent parameters. The GPS module provides accurate geolocation data (latitude and longitude), which improves the overall accuracy of the overall system.

The system is also designed to work in real-time. That is, it will process the input data and produce output instantaneous. All of the calculations are done locally with an edge computing platform; therefore, there is no requirement for an active Internet connection, which helps to minimize latency. The TFT touchscreen continuously updates the information provided to the user; thus providing the user with up-to-date, easy-to-understand results in real-time.

Another major advantage of this system is its power consumption, which makes it feasible to operate off of batteries. Its compact size and portability allow it to be used in virtually any location without modification or alteration to the system. The experiment's outcome demonstrates that this system is reliable and highly accurate and achieves uniformly consistent operating attributes under varying user conditions using the combined power of RTC, GPS, and implemented optimization techniques. The results produced by the system are virtually identical to the standard Panchangam result with only minor deviations from the standard. As these differences are not significant, they will have no effect on the overall quality of the system.

Another major advantage of this new system is that it does not require an internet connection. Therefore, it can be used in remote and rural areas. Its portability also adds to its usefulness. The system allows users to use advanced technology to replicate the calculations of ancient astrologers. The system supports the calculation of panchangams in less complicated ways, while making the results presented in a more simplified manner to be easily understood by all users. There are many other components that can be added to the system (in addition to RTCs and GPS) to improve system functionality and demonstrate that there are still opportunities to extend into many new areas (such as environmental monitoring).

The performance of the Device was assessed across many different parameters to verify its ability to deliver stable, consistent output. The Arduino Nano processed all calculations (solar and astrological), and could perform these calculations at a rate sufficient to be used in real-time applications, even with limited resources. The RTC module supports accurate timing, and the GPS module provided precise location-based calculations.

One of the key features that make this system valuable is the ability to operate in real-time due to the fact that the system continually processes incoming data and updates the display immediately. Because all of the processing is done locally (edge computing), the system can avoid latency that results from being dependent upon a network. The TFT display will dynamically refresh the output to provide users with the most up-to-date information available.

The Device has exhibited high reliability (consistently) over numerous test runs. Low power consumption and portability make this system practical for a wide range of daily uses. In summary, this system provides an efficient, accurate, and easy-to-use solution for astrological numerical computation.

5.3 CONCLUSION

This chapter provides an overview of the electronic astrology calendar and results based on the data discussed earlier. The results confirm the effectiveness of the electronic astrology calendar in providing timely and accurate astrological calculations. The data from the electronic astrology calendar may be compared to the standards provided by the Panchangam; these results are all considered acceptable due to small differences. The small differences noted above do not impede or restrict the practical usefulness of the electronic astrology calendar.

The electronic astrological calendar system meets all of the designed requirements outlined prior to this chapter such as being portable, having the ability to work offline, along with being user-friendly. The technology used in the electronic astrological calendar system is embedded system technology, which allows for all calculations to be calculated at the local level. Therefore, the speed in executing the calculations from the electronic astrological calendar system will not experience delay. The use of low power components ensures that the electronic astrological calendar system is energy-efficient and can operate effectively on battery power.

The electronic astrological calendar system provides users with a user-friendly interface to see their results and will allow users to easily interpret their results. Overall, the electronic astrological calendar system will provide the user with a practical, efficient, and modern means of obtaining astrological information; therefore, successfully combining the traditional knowledge of astrology with the latest technology advancements.

The electronic astrological calender system provides a TFT (thin-film transistor) display, which is a user-friendly graphical presentation of astrological information. The graphical display provides easy-to-understand data to the user, allowing even those with minimal technical expertise to interpret the data. The software also automatically updates as information is changed, so there is no need for users to input data manually. Therefore, the astrological calendar system is an effective way to obtain astrological information, providing a practical, efficient, and modern method of accessing astrological information based on both traditional knowledge of astrology and new technological advancements. Additionally, the reliability and accuracy of the overall system means it can be used in a wide range of settings, including locations where no Internet service is available (e.g., the most remote areas). In summary, the astrological calendar system is a valuable resource for accessing astrological information.

CHAPTER-6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

The proposed system, "Edge AI-Based Electronic Indian Astrological Calendar," has been successfully designed and implemented by utilizing the Arduino Nano platform. The hardware components used in this project, like the RTC module (DS3231), Neo-6M GPS module, BMP280 sensor, and 2.4-inch TFT display, have been integrated to develop a complete hardware solution to compute and display the astrological details. The design and development of this project have been greatly influenced by the literature survey conducted, which helped to develop a strong foundation in the design of embedded systems and astronomical calculation methodologies.

The computed values have been observed to be quite accurate, with minimal deviations, when compared to the values obtained from the panchangam sources.

In conclusion, it can be said that the proposed system is a reliable, accurate, and user-friendly system for accessing daily astrological information. The system has eliminated the limitations of traditional and internet-based methods of accessing astrological information. The project has successfully demonstrated the potential of embedded technology and edge computing in the creation of practical applications, thus making it a successful contribution in the field of technology as well as ancient astrological practices.

6.2 FUTURE SCOPE OF THE WORK

The proposed edge AI-driven electronic Indian astrological calendar has been successful at providing accurate and reliable information in an offline capability. However, there is still much room for improvement in all aspects of functionality, accuracy, and user experience. Some areas that could benefit from improvement include expanding on the types of astrological parameters calculated by the system. Currently, the astrological parameters calculated by the system are limited to sunrise, sunset, Rahu Kalam, Yamagandam, and Gulikai; therefore, additional calculations would make the efficiency and accuracy of the system much greater. In the future, the accuracy of calculating Panchangam details (i.e., Nakshatra, Tithi, Yoga, and Karana) as well as additional detail regarding the position of planets will also enhance the overall comparison of this to a digital Panchangam system.

The addition of wireless communication technology such as Wi-Fi and Bluetooth presents another opportunity for the system to be extended. This extends to adding wireless connectivity to the system for synchronizing data to other mobile applications or devices. Synchronization would be accomplished by connecting the system to mobile application(s) designed to work in conjunction with the system. Even though the current system is designed for operation in an offline mode, there is room for enhancement by making these features optional, along with the ability to add new functionality.

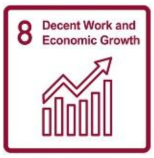
The implementation of state-of-the-art edge AI technology should allow for significant increases in system intelligence. Future generations of this device could provide users with personalized astrological consultations based on user's selections, prior experiences, and/or typical usage patterns. For instance, it could provide suggestions for optimal timing for particular activities, as well as alert users of significant events. Conclusively, the machine learning modules running on edge devices will collect and analyze data points for pattern recognition which will improve the accuracy of predictions as more data is collected over time. Ultimately, the end user will be able to take the benefit of an intelligent system in the form of a virtual personal assistant who will enable them to obtain quality information relevant to their everyday needs.

GPS modules could also be further developed for maximum available efficiency. Effective ways to enhance GPS module functionality include increasing the available speed of signal acquisition, as well as increasing the available degree of accuracy associated with location determination.

SUSTAINABLE DEVELOPMENT GOALS

The following Sustainable Development Goals (SDG) are covered by the proposed project work.

Table SDG Proposed Project SDG Coverage Table

SDG SYMBOL	SDG DESCRIPTION	PROPOSED PROJECT
	Affordable Clean Energy	The system is designed using low-power components like Arduino and RTC modules, making it energy-efficient. This supports sustainable energy usage and reduces power consumption.
	Decent Work and Economic Growth	This project can create opportunities in embedded systems development, IoT-based products, and traditional digital services. It can also support small businesses or temples by providing a low-cost astrological solution.

	Industry innovation and Infrastructure	The project uses **Edge AI and embedded technology**, showing innovation in combining traditional astrology with modern electronics. It contributes to technological advancement and smart device development.
	Sustainable Cities and Communities	The device is portable, reliable, and can be used in temples, homes, or public places. It supports smart and culturally connected communities by digitizing traditional systems.
	Responsible for production and consumption.	The use of compact hardware, minimal resources, and efficient programming ensures reduced waste and sustainable product design.

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EDGE - AI DRIVEN ELECTRONIC INDIAN ASTROLOGICAL CALENDAR FOR OFFLINE APPLICATION

A PROJECT REPORT

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in partial fulfilment for the award of the degree

of

BACHELOR OF ENGINEERING

in

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Dyanesh M **Raghav Kanva V R** **Soundrarajan S A**

ABSTRACT

Using Indian Astrology, the Project will provide an offline-edge AI based application that offers daily yoga times (sunrise/sunset, Rahu Kalam, Yamagandha, Gulika, etc.) via pre-programmed algorithms that are relative to the user's location in conjunction with their dates/times as stored within a Real Time Clock (RTC) using the Arduino microcontroller to ensure date/time activities; astrological data will be derived quickly and accurately without manual calculations or requiring references over the Internet.

A BMP280 sensor will measure environmental variables, including temperature and pressure, etc. The UTC generated from these variables will then be displayed on an LCD screen for easier reading and usability. The rooted compact size of this system (lightweight) can enhance its portability and usefulness in rural or isolated areas. Additionally, Edge-AI technology will allow the project to be much more efficient, accurate, and reliable than traditional ways to validate the Panchangam while no longer requiring Cloud Services for obtaining current astrological data.

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LIST OF ABBREVIATIONS

GPIO	General Purpose Input Output
USB	Universal Serial Bus
AI	Artificial Intelligence
RTC	Real-Time Clock
TFT	Thin Flim Transistor
GPS	Global Positioning System
BMP	Barometric Pressure Module
SDA	Serial Data Line
SCL	Serial Clock Line
LPES	Low Powered Embedded System
PCB	Printed Circuit Board